

Satellite Communications

ECE 4644 / 5610 Spring 2026

Week #6

Feb 23 - 27, 2026

Instructor: J. Michael Ruohoniemi

Topics (1/2)

- Antennas (continued)
- Reliability and Space Qualification
- Link equation
- **Mid-term prep**
- Noise power / Noise models / Noise Figure
- Carrier-to-Noise ratio, G_r / T_s ratio

Topics (2/2)

- Atmospheric and rain attenuation
- C/N ratio on satellite links – uplink, downlink and overall C/N

SES-14 Satellite (47.5° W)

- Communication satellite in GEO orbit positioned over the Atlantic ocean
- Launched in 2018 by Ariane
- Replaced an older satellite operating in C-band
- Includes a C-band payload and a Ku-band payload

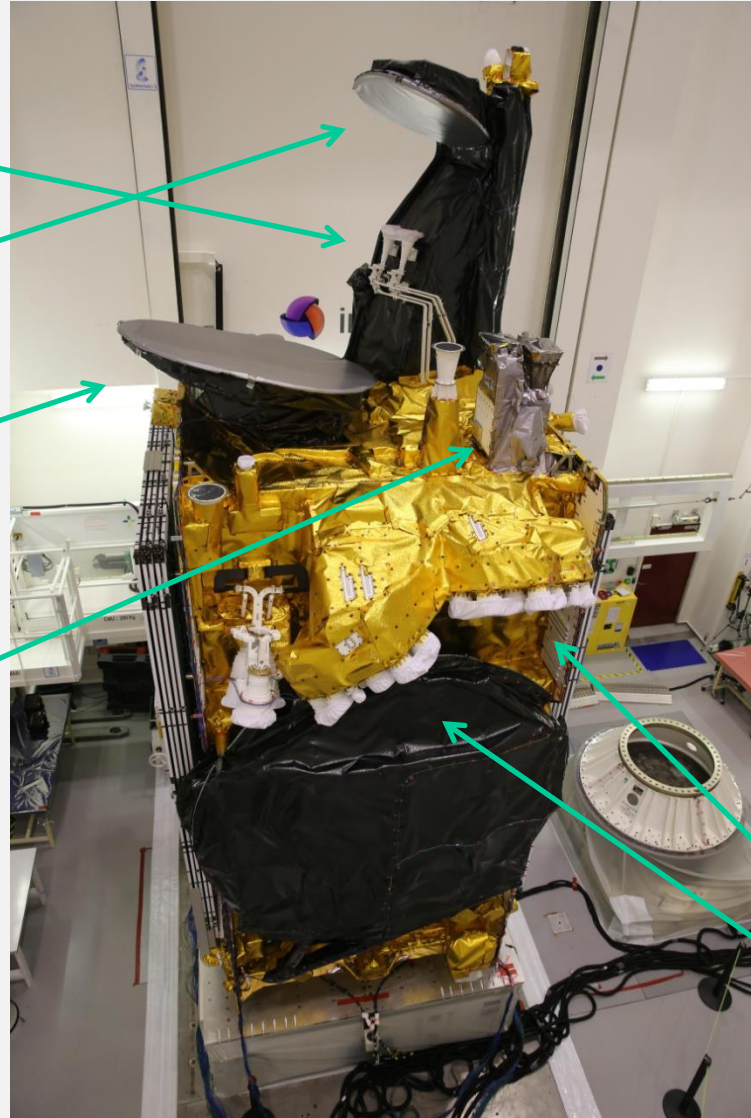
SES-14 Satellite

Feed horns

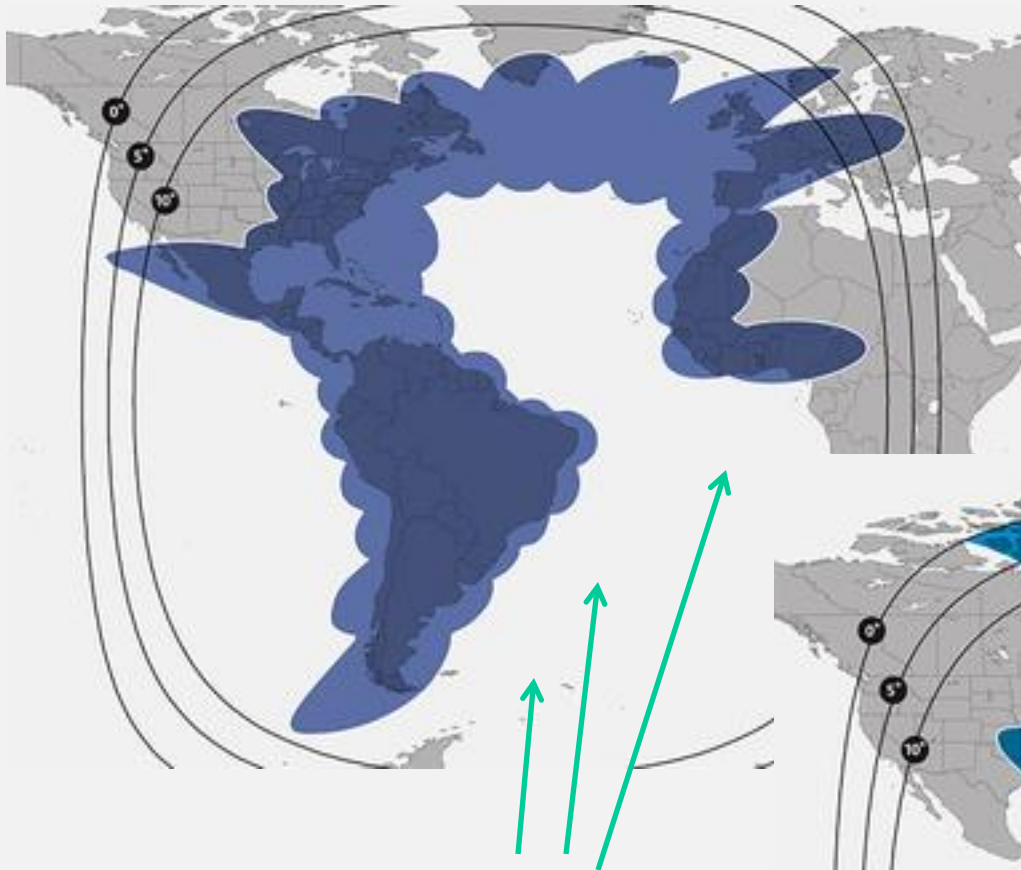
Small dish antenna

Large dish antenna

Includes a NASA
scientific instrument
package: Global-
scale Observations
of the Limb and
Disk (GOLD)



???



SES-14 Ku-band
coverage

Ku-band wide beams



Ku band spot
beams (requires
multiple horn
feeds)

Earth Station Antennas

- Hub antennas are large, fully steerable, and very expensive - \$ 5M
 - > high capacity, multi-channel, high revenue applications
- Consumer antennas need to be very small and low cost – a few hundred dollars
 - > Direct Broadcast Satellite TV

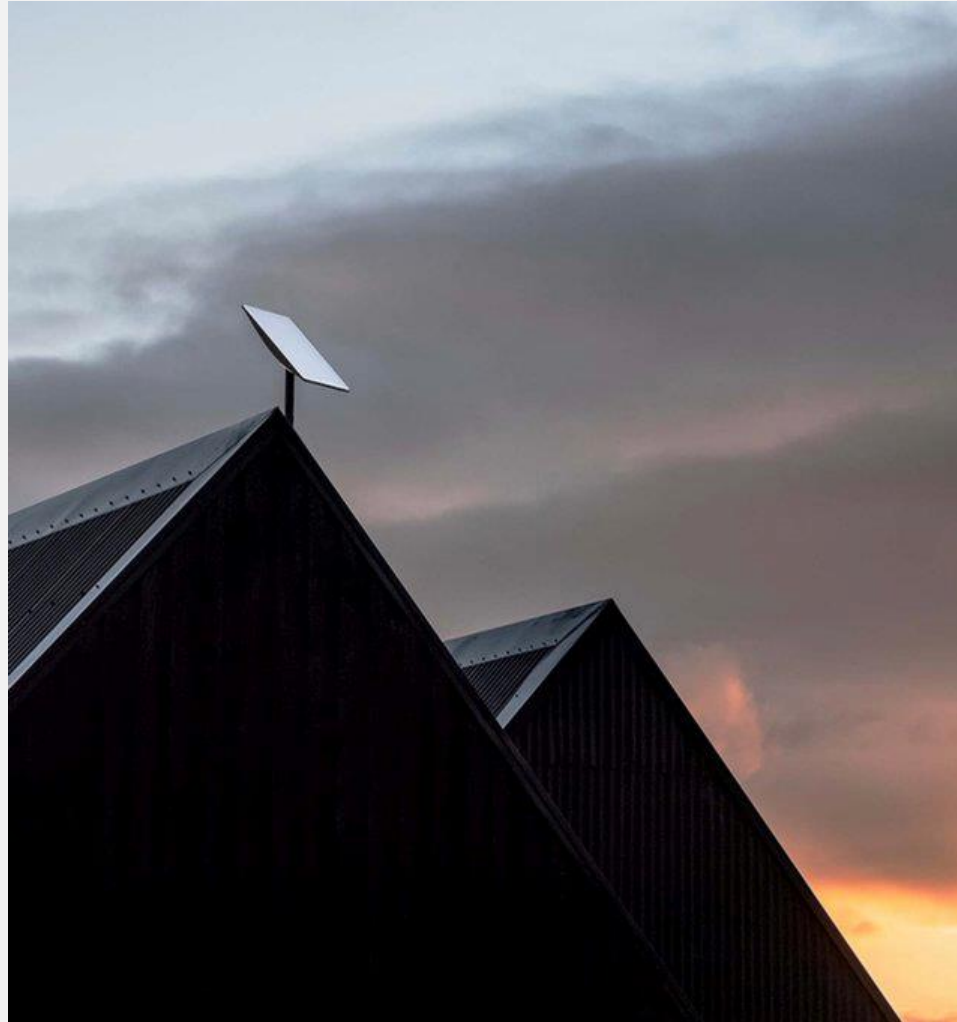


Fig 11.4

Megaconstellation Consumer Antennas

- Satellites in LEO orbit need to be tracked with phased array antennas
- A major challenge for Starlink and other companies was to make the consumer phased array antenna small and affordable
- Starlink basic consumer phased-array antenna costs \$499 to buy and \$1000 to manufacture

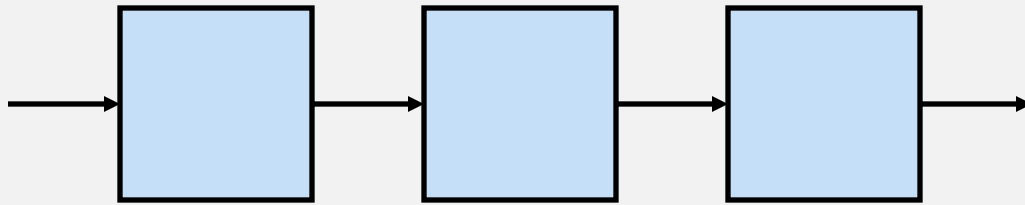
Starlink rectangular phased-array antenna



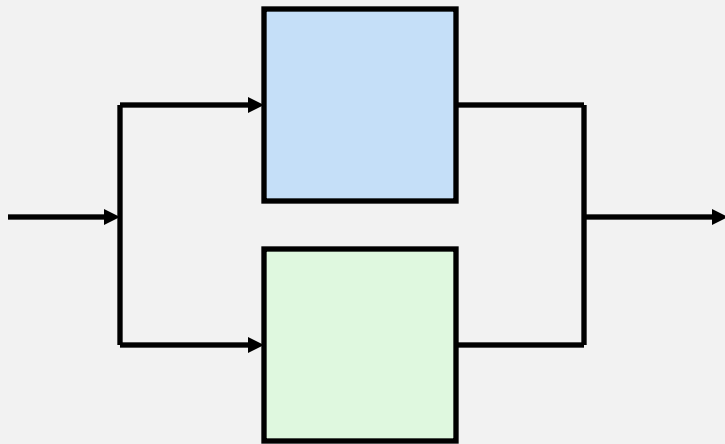
Reliability

- Satellites cannot be easily repaired in orbit
- Reliability and lifetime are important issues
- Some components have limited lifetime
- E.g. High power amplifiers (HPAs)
- *Redundancy* is used to overcome failures
- E.g. HPA may have two amplifiers in parallel
- If one fails the other continues to operate

Redundancy configurations

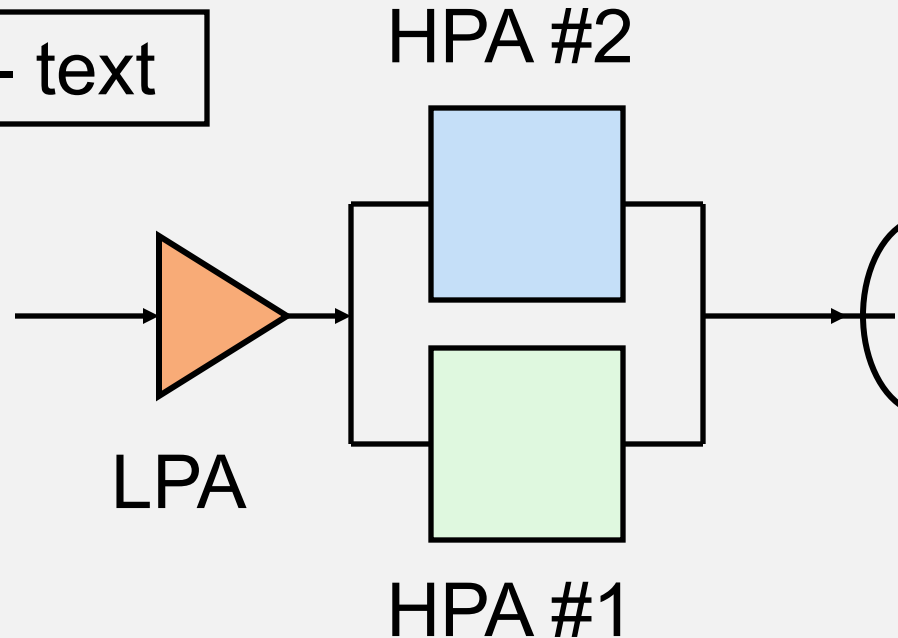


Series redundancy – blocks must fail short circuit



Parallel redundancy – blocks must fail open circuit

Part Fig 3.18 - text



One HPA #2 is off initially, HPA #1 is on.
When #1 fails, #2 is turned on.

Redundant HPAs in output stage of transponder

Components are subjected to a burn-in period before installation

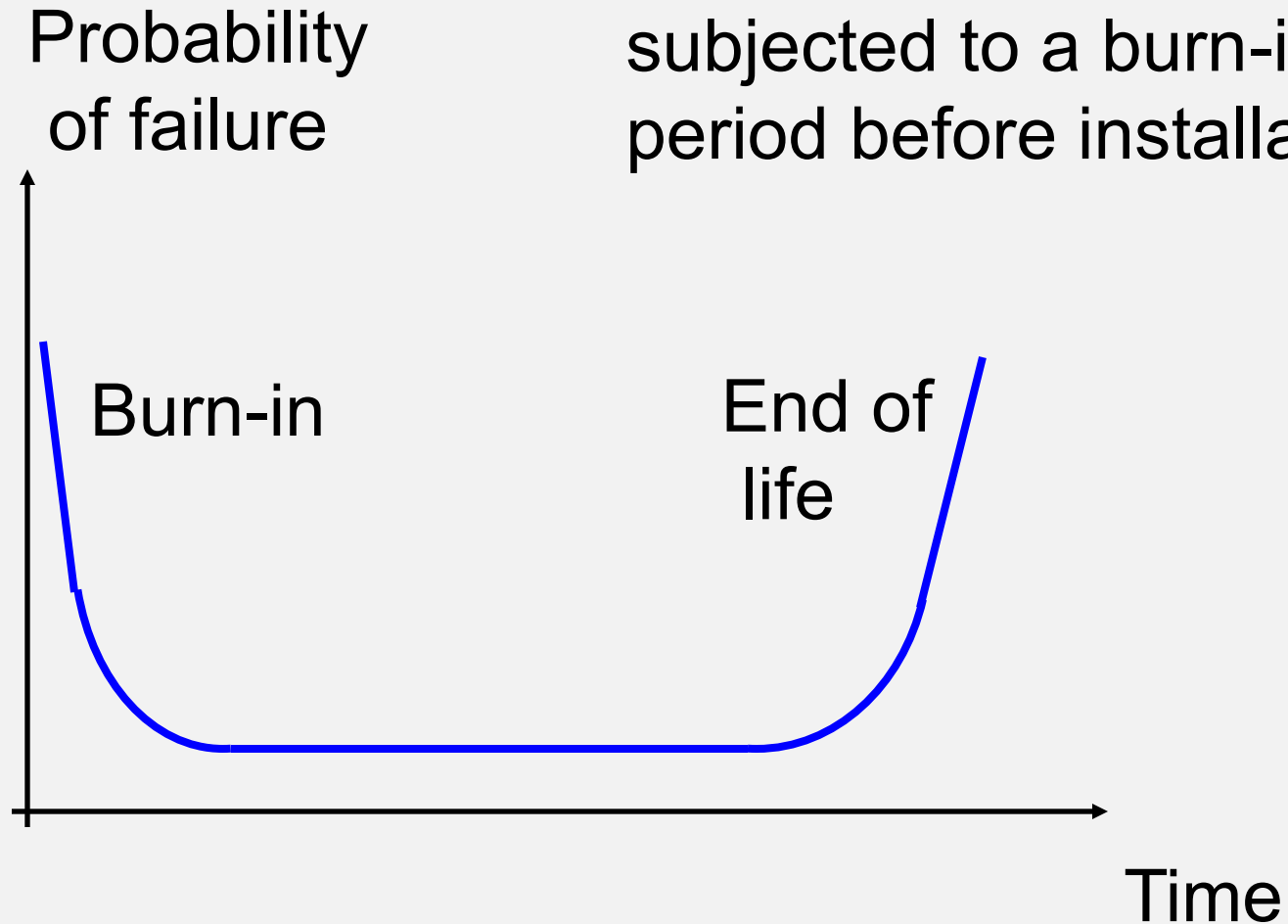


Fig 3.17 - text

Bathtub curve for reliability

Space Qualification

- Reliability can be improved by *space qualification*
- Space qualification involves:
 - selection of high reliability components
 - rigorous testing of all parts and assemblies
 - *burn in* to weed out early failures
 - *shake and bake* tests

Space Qualification

- A satellite will experience temperature extremes ($-100\text{ C} > 100\text{ C}$)
- Thermal control system is needed to manage heat flows
- A satellite will experience vibrational stress through launch
- Thermal, vacuum, and vibration testing is performed on the entire satellite in a thermal vacuum chamber in the shake-and-bake test

Summary

- Satellite has many subsystems
- Power, antennas, and communications are important for this course
- Transponders handle traffic and earn revenue
- Antennas must provide correct footprints
- High reliability is essential
- Critical components are supported with redundancy

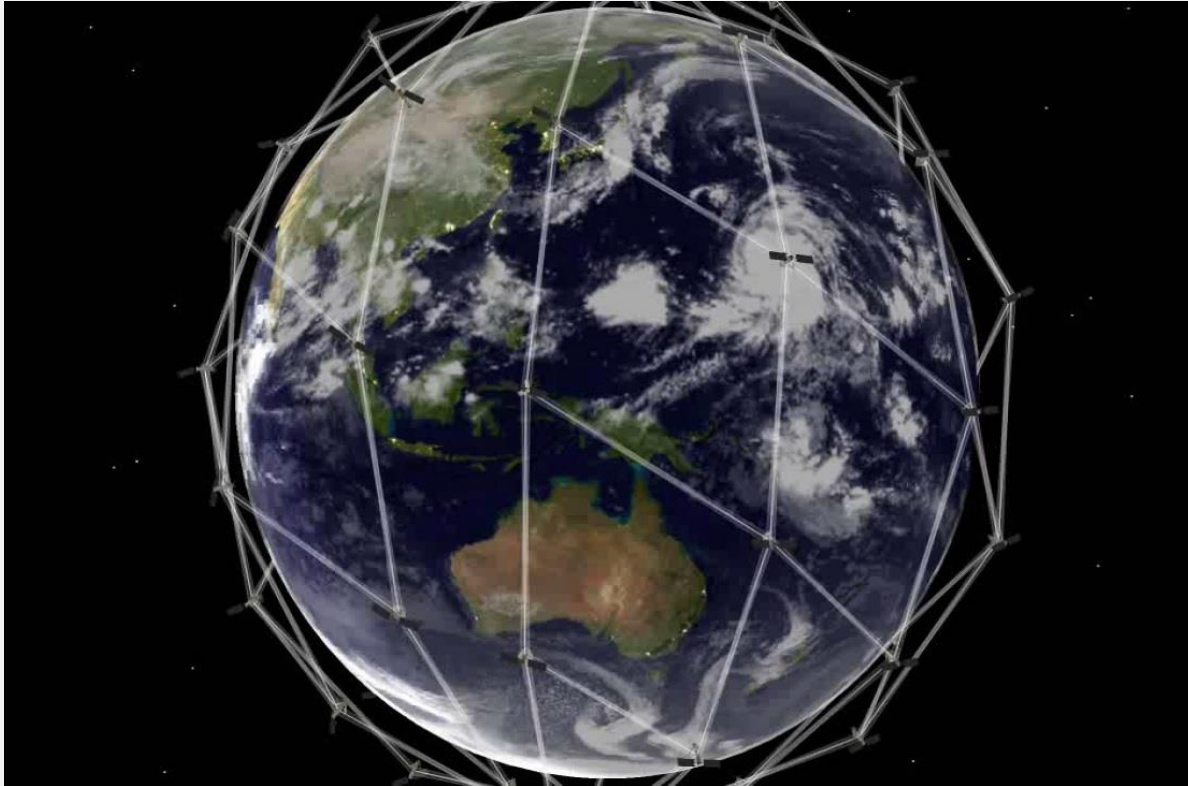
Frequency Bands

- Satellite communication systems must operate in allocated frequency bands
- The International Telecommunications Union (ITU) Radio group (ITU-R) allocates frequencies at World Radio Conferences (WRCs)
- In the US, the Federal Communications Commission (FCC) manages use of the (civil) radio spectrum

Frequency Bands

- Major bands for SatComm are the 6/4, 14/11, and 30/20 bands
- Uplink frequency is quoted first (in GHz)
- Correspond to the C, Ku, and Ka bands
- V-band, 40 - 75 GHz, supports large constellations of satellites in LEO primarily for *inter-satellite links* (ISLs)

Iridium-NEXT: Ka-band Inter-satellite Links



Each satellite connects with up to four neighbors – two in the same orbital plane and two in adjacent planes

Frequency Bands

- L-band:
1600 / 1500 MHz Mobile Satellite Systems (MSS)
 - Uplinks 1625 – 1660 MHz
 - Downlinks 1530 – 1559 MHz
- C-band (original allocations):
6/4 GHz Fixed Satellite Service (FSS)
 - Uplinks 5795 – 7075 MHz
 - Downlinks 3400 – 4200 MHz

Frequency Bands

- Ku-band:

14/11 GHz Fixed Satellite Service (FSS)

- Uplinks: 14.0 – 14.8 GHz
- Downlinks 10.7 – 11.7 GHz

- Ku-band:

Direct Broadcast Satellite Television (DBS)

- Uplinks 17.3 – 17.7 GHz
- Downlinks 12.2 – 12.7 GHz

Frequency Bands

- Ka-band:

30/20 GHz Fixed Satellite Service (FSS)

- Uplinks: 27.5 – 29.5 GHz
- Uplinks: 30.0 – 31.0 GHz
- Downlinks 17.7 – 19.7 GHz

- V-band:

50/40 GHz Constellations of satellites in LEO

- See Tables 4.1 and 4.2 in class text for details

Frequency Bands - Military

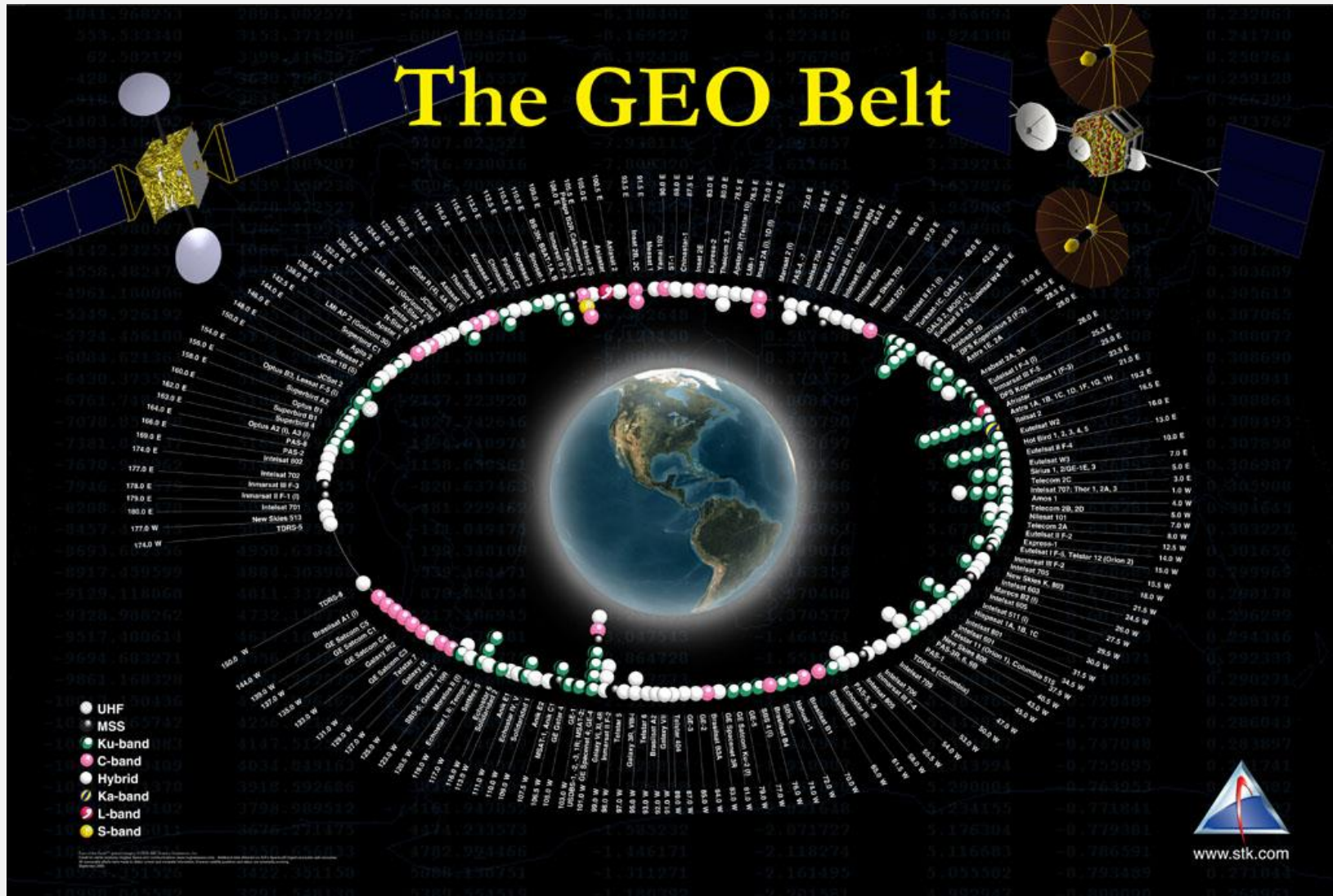
- C-band

- 8 / 7 GHz satellite links – DSCS

- Uplinks: 8.215 – 8.40 GHz
 - Downlinks 7.25 – 7.75 GHz

- 44 / 22 GHz satellite links – Milstar

- Uplinks: 44 GHz
 - Downlinks 20 GHz
 - Crosslinks 60 GHz



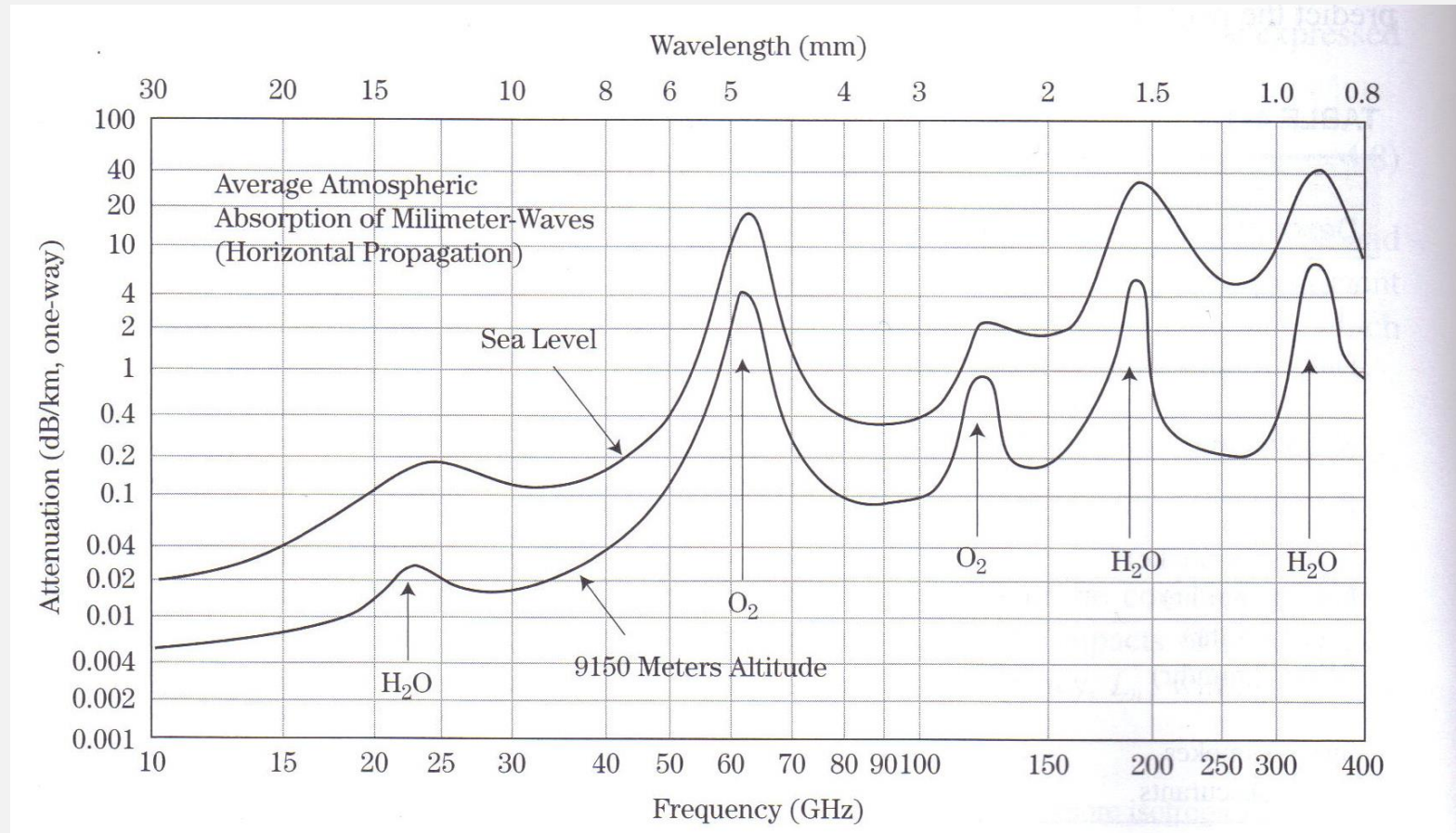
Selecting a frequency

- Must be within allocated bands
- Satellites in GEO can share a slot by operating on different bands
- GEO orbit filled at C-band, Ku band in most slots (2° spacing, populated areas)
- May have filled with Ka-band satellites
- The problem with raising frequency:
atmospheric loss and weather (rain) loss increase

Propagation in earth's atmosphere

- Earth's atmosphere is a 'lossy' medium to RF signal
- Satcomm will suffer attenuation due to absorption by neutral gases in the atmosphere and by rain
- Both types of loss generally increase with frequency in the GHz range
- Must be factored into the design of a communications system

Attenuation Coefficient due to Atmospheric Gases



From *Principles of Modern Radar*, Richards et al., 2010

Propagation in earth's atmosphere

- Attenuation in clear air
 - Atmospheric gases: Water vapor, Oxygen
 - Water vapor absorption 22 – 23 GHz
 - Oxygen resonance 55 – 60 GHz
- Attenuation in rain
 - Not very significant below 10 GHz
 - Increases approx as frequency squared

Propagation in earth's atmosphere

- Satellite communication links must be designed to allow for atmospheric attenuation
- Two parts: clear air, rain
- Clear air has atmospheric attenuation – even when it rains
- Allow 0.2 dB at C band, 0.5 dB at Ku band, 1 dB at Ka band
- When rain occurs in a satellite link it causes attenuation

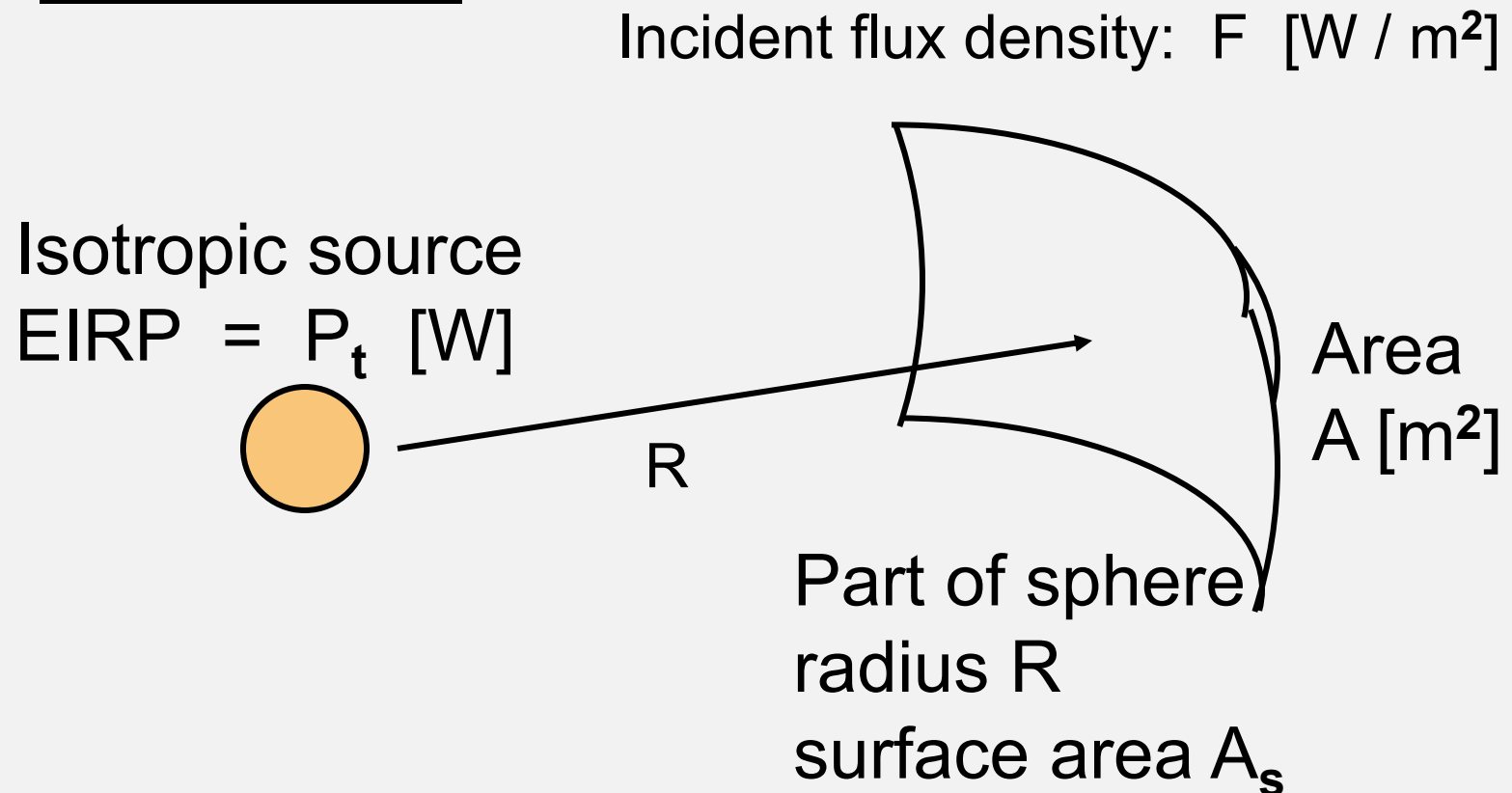
Propagation in earth's atmosphere

- Each satellite link is designed to withstand a specific rain attenuation
- Maximum permitted attenuation is called a *link margin*
- If rain attenuation exceeds the link margin, the link will fail - link is suffers an *outage*
- Design must be based on rainfall statistic

Link Equation

- We need a means to calculate the received power, P_r [watts], in a radio link
- The link equation solves for P_r in terms of the transmitted power, P_t [watts]
- Parameters are:
 - Transmitted power, P_t
 - Antenna gains
 - Distance between transmitter and receiver
 - Radio frequency

Fig. 4.2 - text



Flux density produced by an isotropic source

Flux Density

- Isotropic source with power P_t watts radiates equally in all directions
- Flux density (or power density) at distance R meters is F W/m²
- F is radiated power divided by surface area of sphere

$$F = P_t / A_s = P_t / [4 \pi R^2] \text{ W/m}^2 \text{ (Eqn 4.1)}$$

- Flux density is independent of frequency

Received Power

- We need directive antennas in satellite systems
- Antenna has narrow beam, gain G (a ratio)
- Gain describes the ability of an antenna to increase power transmitted in a particular direction
- We can combine gain and transmitted power:
$$\text{EIRP} = P_t G_t \text{ watts} \quad (\text{sometimes ERP})$$
- EIRP = Effective Isotropically Radiated Power

Antennas (Reminder)

Definition of antenna gain:

The increase in received power at a given point with the test antenna relative to the power received from an isotropic antenna

Definition of an isotropic antenna:

An antenna that radiates equally in all directions - does not exist

Received Power

- For a source with EIRP = $P_t G_t$ watts

$$F = P_t G_t / [4 \pi R^2] \text{ W/m}^2 \quad (\text{Eqn 4.3})$$

- Power received by an aperture with a physical area A_p [m²] is

$$P_r = F \times A_e = F \times \eta_A A_p \text{ W} \quad (\text{Eqn 4.4})$$

where A_e is the *effective area* of the aperture and η_A is the *aperture efficiency*

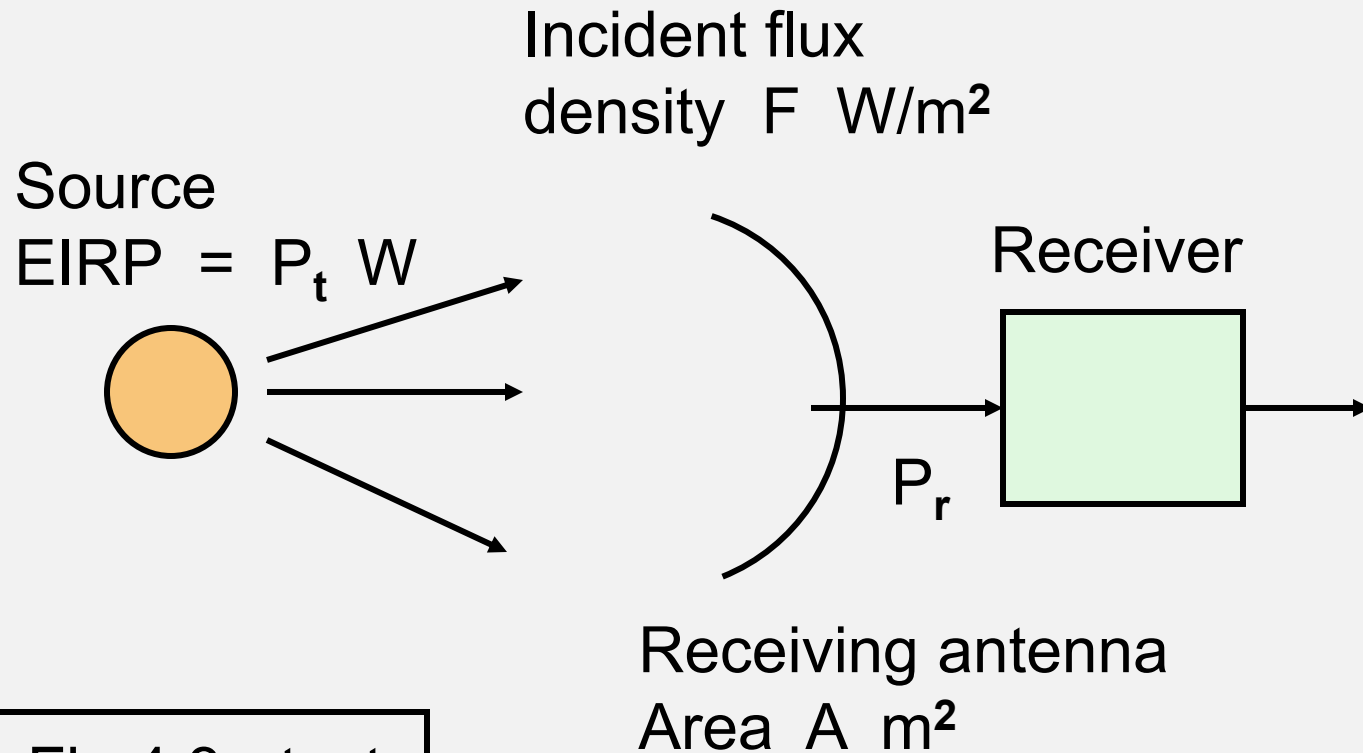


Fig 4.3 - text

Received Power

- From antenna theory, the gain of an antenna is related to its effective aperture by

$$G = 4 \pi A_e / \lambda^2 \quad (\text{Eqn 4.7})$$

- Hence

$$A_e = G_r \lambda^2 / 4 \pi$$

$$P_r = F \times A_e = P_t G_t G_r / [4 \pi R / \lambda]^2 \quad W$$

(Eqn. 4.8)

- This is the basic link equation (no losses)

Received Power

- The term $L_p = [4 \pi R / \lambda]^2$ is called the *path loss*
- It is not a loss in the sense of power being absorbed
- It describes how power spreads out with distance, giving loss proportional to $1 / R^2$
- Engineers prefer to think in terms of *gains* and *losses*

Received Power

- Link equation in terms of EIRP: (Eqn 4.9)

$$P_r = \frac{\text{EIRP} \times \text{Receive antenna gain}}{\text{Path loss}} \quad \text{W}$$

- Link equation in decibel units:

$$P_r = P_t + G_t + G_r - 10 \log [(4 \pi R) / \lambda]^2 \quad \text{dBW}$$

where power and gain terms are in dB units

Link Equation in dB

- Communications engineers make calculations in decibels
- Define dB:

$$\text{Value in decibels} = 10 \log_{10}(P_2 / P_1)$$

- The decibel is defined as the ratio between two powers – not two voltages
- Don't use $20 \log_{10}()$ unless you are calculating path loss and taking the square outside the log

Using dB

- To multiply: Add in dB
- Example:

$$\begin{aligned} P_t G_t &= \text{transmit power} \times \text{antenna gain} \\ &= \text{EIRP} = \text{effective isotropic radiated} \\ &\quad \text{power} \\ &= P_t \text{ in dBW} + G_t \text{ in dB} \end{aligned}$$

Using dB

- To multiply: Add in dB
- Example:

$$P_t = 20 \text{ W} = 13 \text{ dBW}$$

$$G_t = 100 = 20 \text{ dB}$$

$$\text{EIRP} = 20 \times 100 = 2000 \text{ W} \quad (\text{linear})$$

$$\text{or} \quad = 13 + 20 = 33 \text{ dBW} \quad (\text{dB units})$$

Using dB

- To divide: Subtract in dB
- Example:

$$\text{Path loss} = L_p = (4\pi R / \lambda)^2$$

$$= 10 \log_{10} (4\pi R)^2 - 10 \log_{10} \lambda^2$$

$$= 10 \log_{10} (4\pi)^2 + 10 \log_{10} R^2 - 10 \log_{10} \lambda^2$$

We will use this formula to find path loss later

Received Power

- We will prefer to work the link equation in dB units, i.e.,

$$\begin{aligned} P_r &= P_t + G_t + G_r - 10 \log [(4 \pi R) / \lambda]^2 \\ &= P_t + G_t + G_r - L_p \quad \text{(basic form)} \end{aligned}$$

where $L_p = 10 \log [(4 \pi R) / \lambda]^2$

- This is basic form of the link equation (no losses)

ECE 4644 / 5610 Mid-term (1/3)

- Mid-term will be in-class during the regular lecture time on Wednesday, March 18
- Closed-book; Equation sheet will be provided (and available for review on Canvas website)
- Bring a writing implement and calculator
- Consists of a multiple-choice section, a short problems section, and a section with a longer design-type problem

ECE 4644 Mid-term (2/3)

- Scope: all material to the end of Week #7 (March 4) including all HW (through HW#4 and any demo HW)
- Emphasis will be on understanding key concepts, knowing basic facts, and working problems such as those discussed in class or encountered in homework
- The Monday lecture on March 16 will be prep for the mid-term with review of recent HW and the Equation sheet

ECE 4644 Mid-term (3/3)

- Off-campus students will need to make an arrangement to have the mid-term proctored
- A similar arrangement will be needed for the final exam
- See Announcements

Power versus Noise

- Using the link equation, we can calculate the power available in the receiver from the transmitted signal
- Is this enough to sustain a communications link?
- Depends on the noise in the receiver
- We expect that signal power will have to be higher than noise power for successful performance

Noise

- We need to understand thermal noise
- All electrical circuits with $T > 0^\circ \text{ K}$ have noise
- Random motion of electrons causes fluctuating AC current = *thermal noise*
- Thermal noise sets a lower limit on signal power in any receiver
- We usually need $P_r = N + 10 \text{ dB}$ to make demodulators work correctly in receiver

Noise Power

- A *black body* with a physical temperature, T_p degrees kelvin, generates electrical noise over a wide bandwidth
- The noise power of a black body is given by

$$P_n = k T_p B_n \quad (\text{Eqn 4.13})$$

where

k = Boltzmann's constant

T_p = temperature in kelvin degrees

B_n = noise bandwidth in which the noise power is measured, in hertz

Noise Power

- We will use the idea of a black body to model noise in a receiver
- Each component will be assigned the temperature of the black body that would generate the same amount of noise power
- Note that this is not the physical temperature of the component, and we measure it simply in kelvins (not degrees kelvin)
- Note: k = Boltzmann's constant

$$= 1.39 \times 10^{-23} \text{ J/K} = -228.6 \text{ dBW/K/Hz}$$

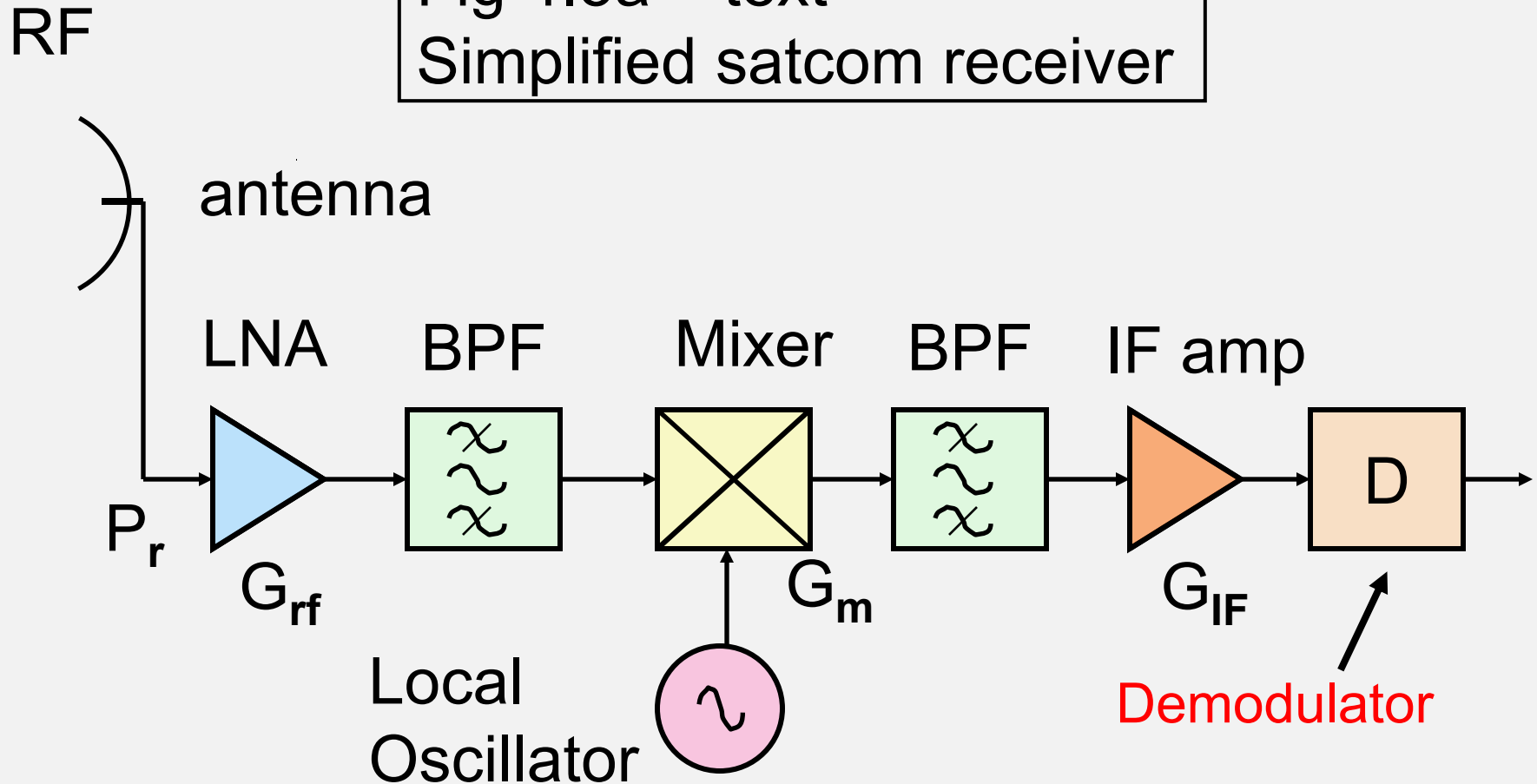
Carrier Wave and Demodulation

- We need to analyze the situation with signal and noise in a receiver
- First, we note that a communication link is sustained by an RF wave that travels from transmitter to receiver
- Sometimes called the *carrier wave*
- Its power is designated by C (compare with N for noise power)

Carrier Wave and Demodulation

- No information is communicated by a pure, unmodulated carrier wave (other than its existence)
- The carrier wave must be modulated in some way with information before transmission
- To extract the information, demodulation is needed at the output of the receiver
- Provided by a *demodulator*

Fig 4.5a – text
Simplified satcom receiver



Note: Carrier wave has been converted from RF to IF

Carrier to Noise Ratio C/N

- Carrier to noise ratio defines performance of system
- We need to know C/N at input to demodulator
- Determines: S/N for analog output
 BER for digital output (error rate)

Noise sources

- Individual blocks in the receiver generate noise
- Thermal noise is present whenever electrons move in a circuit
- We will derive a noise model for receiver:
 - > Same end-to-end gain
 - > All noise in the receiver replaced with a single noise source at the input to the receiver

Model Receiver: in parts

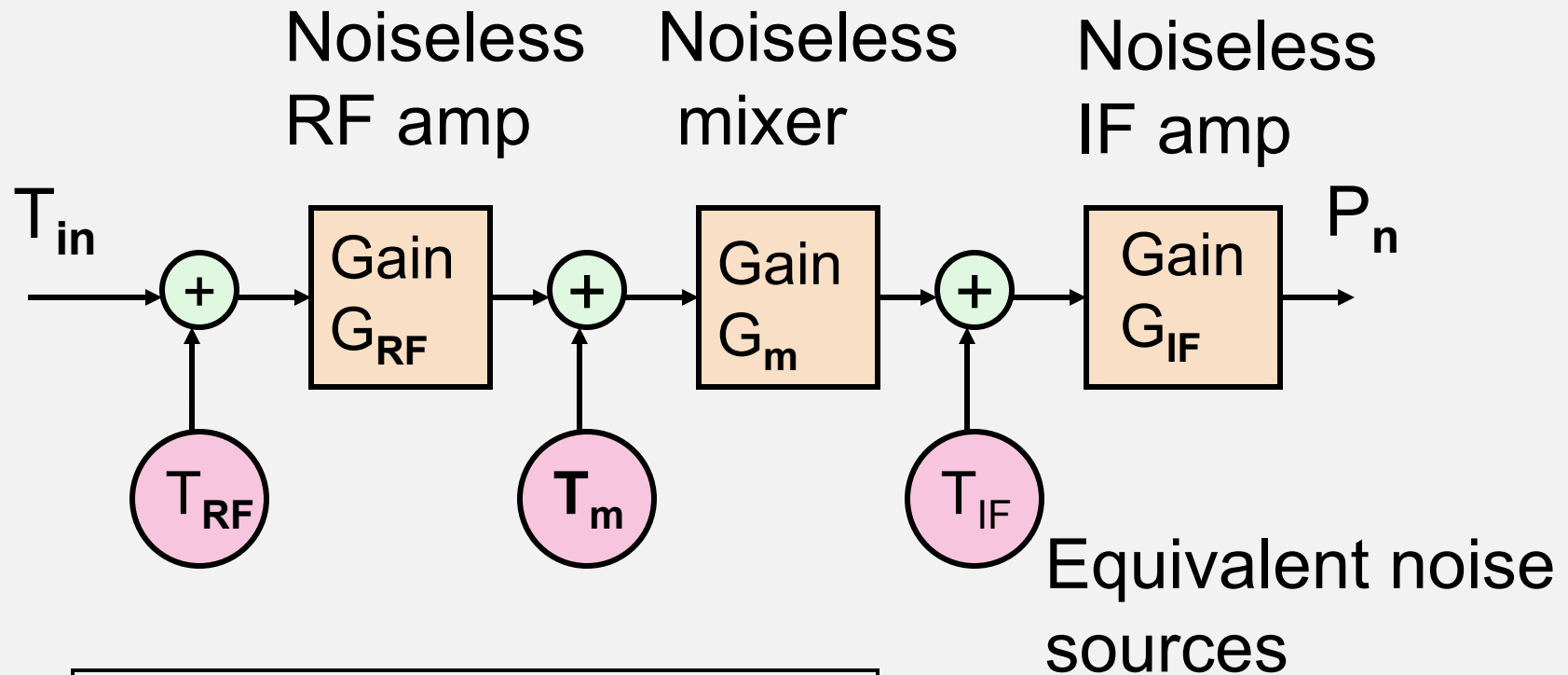


Fig 4.7a - text
Noise model for receiver

Noise Model

- End-to-end gain of receiver is

$$G = G_{\text{RF}} \times G_{\text{m}} \times G_{\text{IF}}$$

- Each noise source contributes noise power

$$N = k T_n B_n$$

$k = 1.38 \times 10^{-23} \text{ J/K}$ Boltzmann's constant

$T_n =$ noise temperature of component

$B_n =$ noise bandwidth of receiver

Model Receiver: in entirety

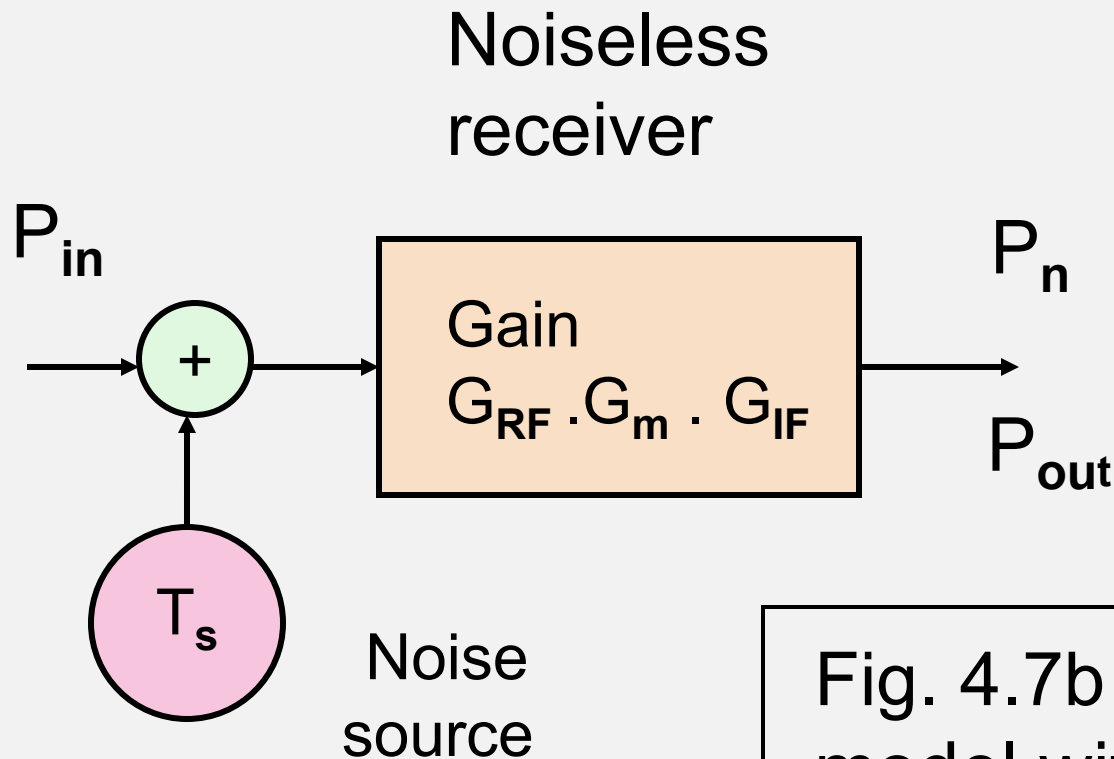


Fig. 4.7b - Noise model with single noise source at T_s

Noise Model for receiver

- Signal Power out of receiver is P_{out}
- Signal Power into receiver is P_{in}

$$\begin{aligned} P_{\text{out}} &= P_{\text{in}} \times G \\ &= P_{\text{in}} \times G_{\text{RF}} \times G_{\text{m}} \times G_{\text{IF}} \end{aligned}$$

- Noise Power out of receiver is P_{n}
- At receiver output (and into demodulator)

$$C / N = P_{\text{out}} / P_{\text{n}}$$

Calculating receiver system noise temperature

- Receiver is modeled as a noiseless receiver with *system noise temperature*, T_s kelvins
- Output of receiver is P_{out}
- Signal : $P_{out} = P_{in} \times G = P_r \times G$
 - Where P_r is the received signal power at the antenna terminals and G is the end-to-end receiver gain
- Noise output from receiver is P_n

Calculating system noise temperature

- $$P_n = G_{IF} k T_{IF} B_n + G_{IF} G_m k T_m B_n + G_{IF} G_m G_{RF} k B_n (T_{RF} + T_{in})$$

[similar to Eqn 4.16]

- G_{RF} , G_m and G_{IF} are the gains of the RF amplifier (LNA), mixer, and IF amplifier
- T_{RF} , T_m and T_{IF} are their equivalent noise temperatures
- T_{in} is the noise temperature of the antenna

Calculating system noise temperature

Noise output of receiver can be expressed as

$$\begin{aligned} P_n &= G_{IF} G_m G_{RF} [(k T_{IF} B_n) / (G_{RF} G_m) \\ &\quad + (k T_m B_n) / G_{RF} + (T_{RF} + T_{in}) B_n] \\ &= G_{IF} G_m G_{RF} k B_n \times \\ &\quad [T_{RF} + T_{in} + T_m / G_{RF} + T_{IF} / (G_{RF} G_m)] \end{aligned}$$

[Eqn (4.17)]

Calculating system noise temperature

Noise output of receiver is

$$\begin{aligned} P_n &= G_{IF} G_m G_{RF} k B_n \times \\ &\quad [T_{RF} + T_{in} + T_m/G_{RF} + T_{IF}/(G_{RF} G_m)] \\ &= G \times k B_n T_s \end{aligned}$$

Hence (Eqn 4.20)

$$T_s = [T_{RF} + T_{in} + T_m/G_{RF} + T_{IF}/(G_{RF} G_m)]$$

We have reduced the receiver noise to a single source at the input to the receiver!

Low Noise Receivers

- Observe that we can minimize noise from receiver by making G_{RF} large
- Example #1: (Ex. 4.3)

$$T_{\text{RF}} = 50 \text{ K} \quad T_{\text{in}} = 25 \text{ K}$$

$$T_{\text{m}} = 500 \text{ K} \quad T_{\text{IF}} = 1000 \text{ K}$$

$$G_{\text{RF}} = 23 \text{ dB} = 200 \quad G_{\text{m}} = 1 = 0 \text{ dB}$$

$$\begin{aligned} T_{\text{s}} &= 50 + 25 + (500/200) + (1000/200) \\ &= \underline{82.5 \text{ K}} > \text{only } 7.5 \text{ K above } T_{\text{RF}} + T_{\text{in}} \end{aligned}$$

Low Noise Receivers

- Example #2: [Later part of Ex 4.3]
- Make G_{RF} larger - e.g., $G_{\text{RF}} = 50 \text{ dB}$

$$T_{\text{RF}} = 50 \text{ K} \quad T_{\text{in}} = 25 \text{ K}$$

$$T_{\text{m}} = 500 \text{ K} \quad T_{\text{IF}} = 1000 \text{ K}$$

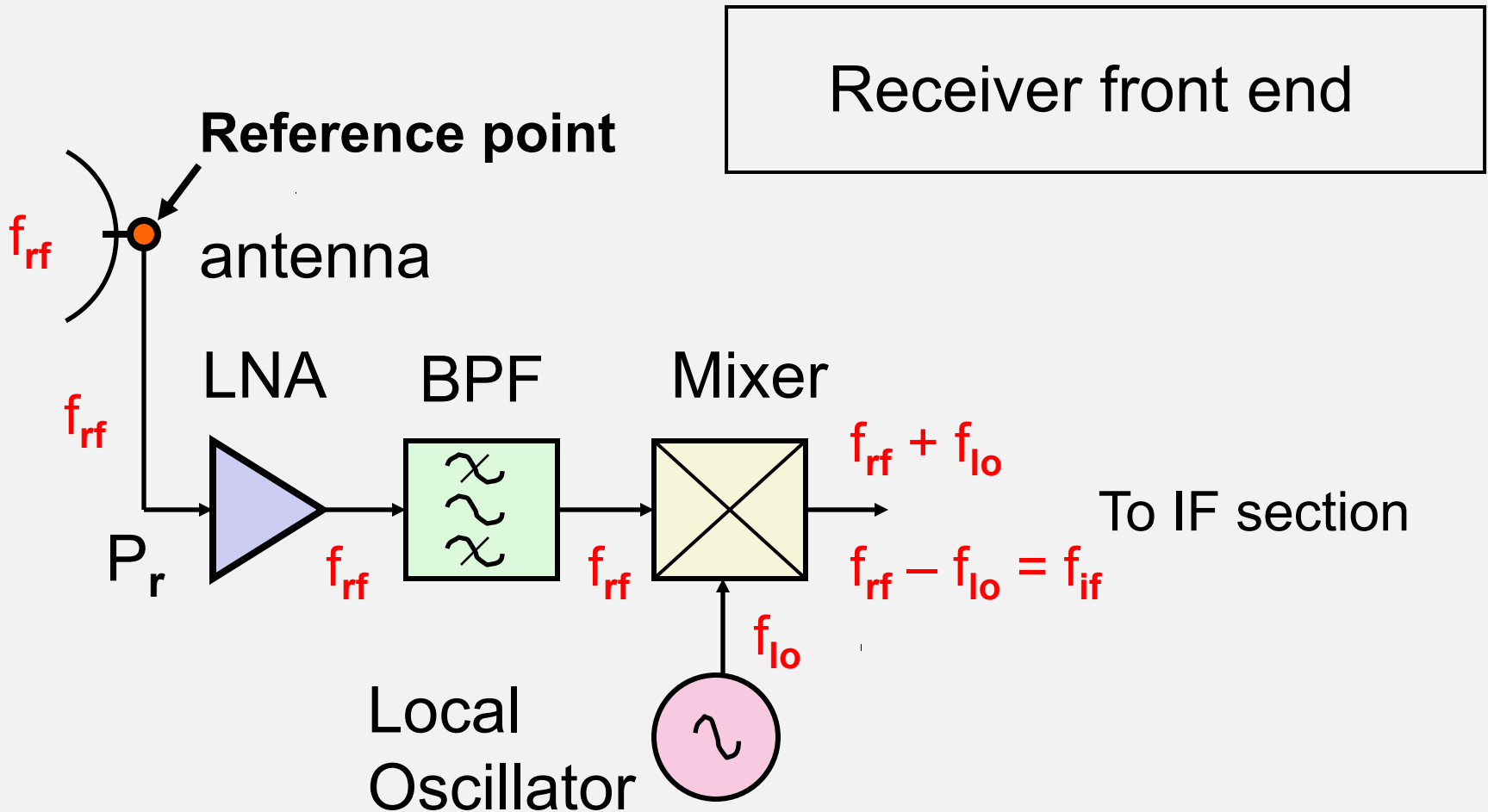
$$G_{\text{RF}} = 50 \text{ dB} = 10^5 \quad G_{\text{m}} = 1 = 0 \text{ dB}$$

$$T_{\text{s}} = 50 + 25 + 0.005 + 0.01 \cong \underline{75.0 \text{ K}}$$

- The components after the RF amplifier now make almost no contribution to T_{s}

Low Noise Receivers

- Conclusions:
 - 1) the RF amplifier should have high gain
 - 2) the RF amplifier should be low noise
> whence its designation as a
Low Noise Amplifier (LNA) <
 - 3) the RF amplifier should be mounted as close to the Rx antenna outputs as possible



Reference point for T_s : output terminals of the antenna

Noise Model

- Recall that a noise source contributes noise power

$$N = k T_n B_n$$

$k = 1.38 \times 10^{-23}$ J/K Boltzmann's constant

T_n = noise temperature of component

B_n = noise bandwidth of receiver

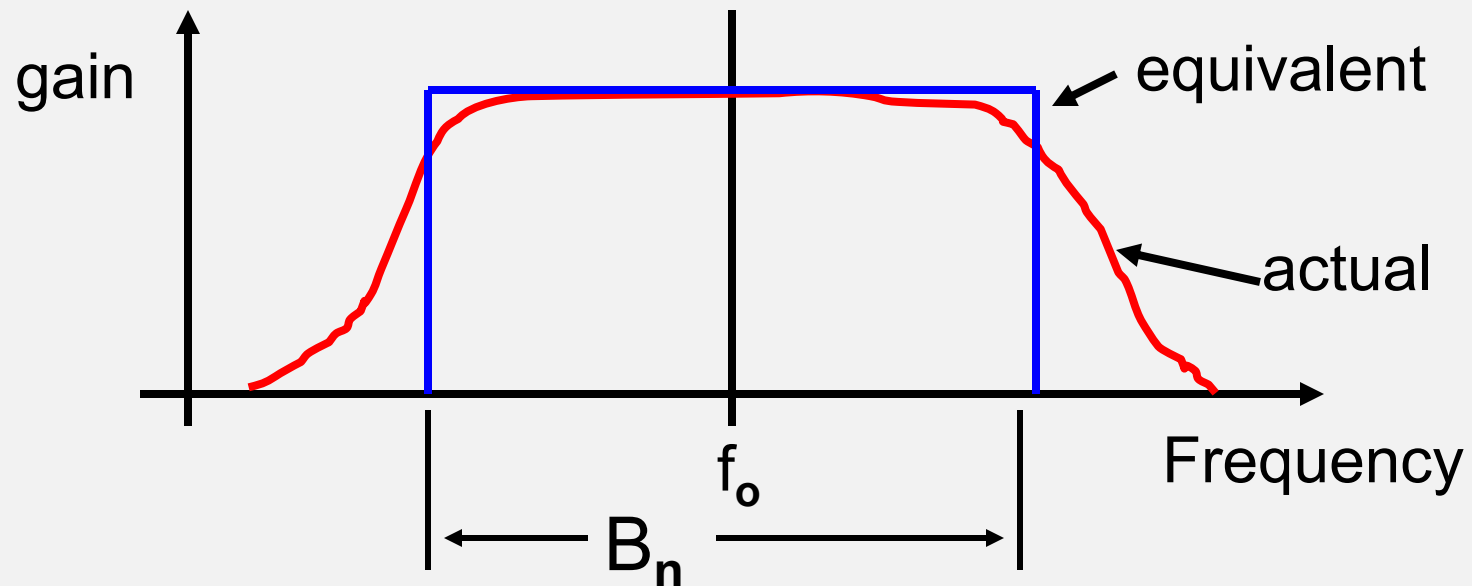
- What exactly is B_n ?

Noise Bandwidth

- The noise bandwidth of a receiver is typically set by a filter
- Noise that is present outside of the passband of the filter does not contribute to the noise power at the output of the receiver
- So, B_n relates to a filter bandwidth

Noise Bandwidth

- Noise bandwidth of a filter is the equivalent rectangular bandwidth that passes the same power as the actual filter



Noise Bandwidth

- All practical bandpass filters set

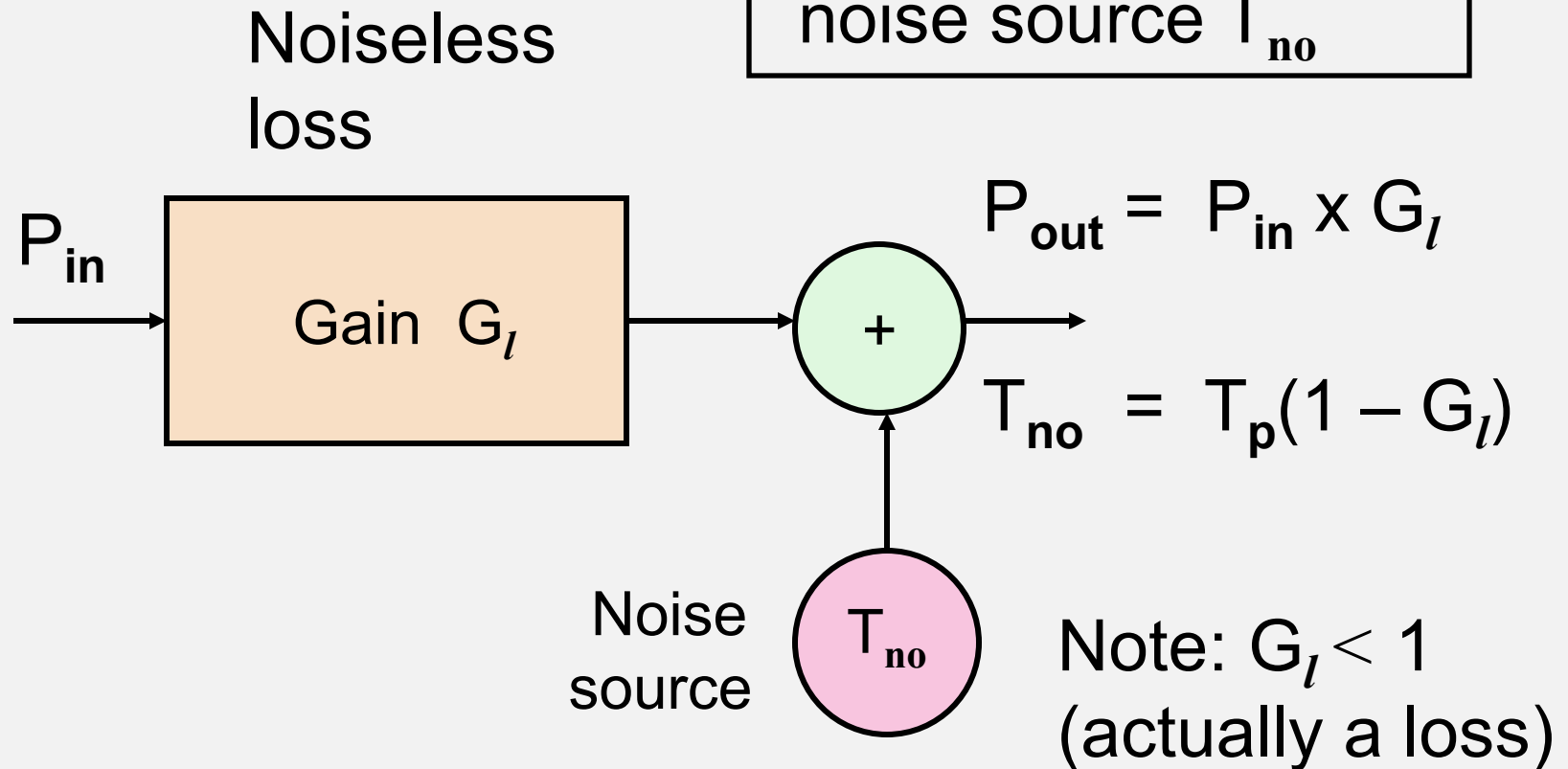
$$B_n = 3 \text{ dB bandwidth}$$

- We use 3 dB bandwidth and call it noise bandwidth
- In many cases, we don't need to calculate T_s
- T_s is often a known value for a given receiver
- E.g. Satcom earth station receivers have system noise temperatures of 50 – 300 K

Noise Model – lossy media

- We need another model to describe noise encountered when signal passes through a lossy medium (waveguide, rain)
- Because a lossy medium radiates noise
- For example, raindrops cause attenuation (absorption) and radiate additional noise
- The level of radiated noise depends on the attenuation

Fig. 4.7c - Output noise model with noise source T_{no}



Noise Models

- *Equivalent output noise source* has noise temperature T_{no} kelvins

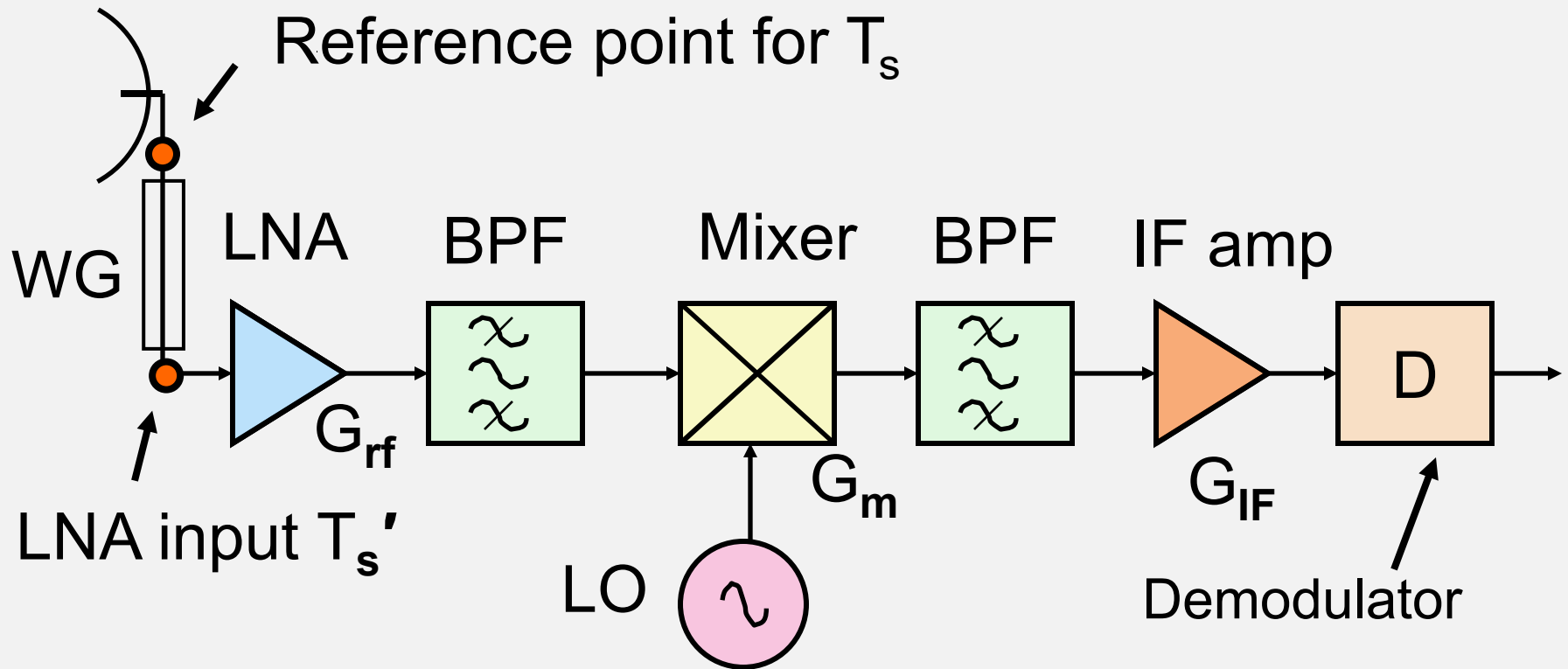
$$T_{no} = T_p (1 - G_l) \quad (4.21)$$

where G_l is the linear gain (less than unity, not in decibels) of the attenuating device

- T_p is the physical temperature of the device

Noisy devices

- Example #3: (after Example 4.4)
- A waveguide with a loss of 2 dB is inserted between the antenna and the LNA in Ex. #2
- The waveguide temperature is 300°K
$$\begin{aligned}T_{\text{in}} &= 25 \text{ K} & T_{\text{RF}} &= 50 \text{ K} \\T_{\text{m}} &= 500 \text{ K} & T_{\text{IF}} &= 2000 \text{ K} \\G_{\text{RF}} &= 50 \text{ dB} = 10^5 & G_{\text{m}} &= 1 = 0 \text{ dB}\end{aligned}$$
- Recall that $T_{\text{s}} \approx T_{\text{RF}} + T_{\text{in}} = 75.0 \text{ K}$ without a lossy waveguide



Calculation of T_s with lossy waveguide inserted before LNA

Noisy devices

- Example #3:
- 2 dB loss means $G_l = 10^{-0.2} = 0.631$
- Equiv. noise temp at output of lossy guide

$$T_{wg} = T_p (1 - G_l)$$

- Using $T_p = 300^\circ \text{ K}$, $T_{wg} = 110.7 \text{ K}$
- This noise appears at the input of the LNA
- And so it adds to the noise power early in the receiver

Noisy devices

- At the input of the LNA we have:
- $T_{\text{RF}} = 50 \text{ K}$
- $T_{\text{wg}} = 110.7 \text{ K}$
- And the noise temperature of the antenna, $T_{\text{in}} = 25 \text{ K}$ *after* it has dropped due to passage through the lossy waveguide

Noisy devices

- At the LNA input, the noise temperature of the antenna has been reduced by 2 dB:

$$T_{in}' = 25 \text{ K} \times 0.631 = 15.8 \text{ K}$$

- System noise temperature at LNA input is then

$$T_s' = 15.8 + 110.7 + 50 = 176.5 \text{ K}$$

- We must transfer the system noise temperature to the antenna terminals where we calculate P_r (this is our reference point)

Noisy devices

- At the antenna terminals, system noise temperature with lossy w.g. is

$$T_s = 176.5 / 0.631 = 279.9 \text{ K}$$

- Originally, T_s was 75 K
- The noise power at the receiver input has increased by 5.7 dB with insertion of only 2 dB of loss ahead of the LNA
- Moral: Keep the loss ahead of the LNA small!

Low Noise Block Converter

- Satellite dishes have the LNA mounted right at the dish, i.e., before the long run of cable into the house
- Forms part of the low noise block converter (LNB) = LNA + mixer + LO + IF amp
- Power comes over the cable from the Rx



Schwaiger LNB2 11.70 - 12.75 GHz Grey Low Noise Block downconverter (LNB)

Noise Figure

- Manufacturers of microwave devices characterize them with a *Noise Figure (NF)*
- $T_n = T_o \times (NF - 1)$
- T_o is the reference temperature (usually 290 K)
- NF is usually given in dB, need to convert to linear to use in this equation
- For satellite communication systems, always convert a noise figure into a noise temperature

Noise Figure

- Step 1.

Convert NF from dB

$$F = 10^{(\text{NF in dB} / 10)}$$

- Step 2.

$$T_n = 290 \times (F - 1)$$

- Example: NF = 2.5 dB

$$F = 10^{0.25} = 1.78$$

$$T_n = 290 \times (1.78 - 1) = 226 \text{ K}$$

Carrier-to-Noise Ratio

- Through the receiver the same gain is applied to the signal power and the noise power
- Meaning that the C/N ratio at the input to the demodulator is the same as at the input to the receiver
- So we can apply the C/N calculated at our reference point (the antenna output terminals) to describe the input to the demodulator

Carrier-to-Noise Ratio

- Apply the link equation with $N = kT_s B_n$ to obtain

$$\begin{aligned} C / N &= [P_t G_t G_r / k T_s B_n] [\lambda / 4 \pi R]^2 \\ &= [P_t G_t] [G_r / T_s] [1 / k B_n] [\lambda / 4 \pi R]^2 \end{aligned}$$

(Eqn. 4.25)

- The first two bracketed terms characterize the situations at the Tx and Rx ends of the link

G/T Ratio

- Transmitting stations are characterized by their $\text{EIRP} = P_t G_t$ [dBW]
- Receiving stations are characterized by their G_r / T_s ratio, usually expressed as G / T in dBK^{-1}
- With $C \propto G_r$ and $N \propto T_s$ we have
 $C / N \propto G_r / T_s$
- EIRP and G / T ratio can be thought of as Tx and Rx performance specs, respectively

Atmospheric attenuation

- The atmosphere attenuates microwave signals
- Rain causes high attenuation on Ku, Ka bands
- Treat atmosphere as a lossy device and calculate output noise
- Noise temperature is called T_{sky}
- Most of the sky noise temperature couples to antenna noise temperature

Atmospheric and rain attenuation

- Example #4
- A Ku-band satellite link has clear air attenuation on the 11 GHz downlink of 0.5 dB
- What is the sky noise temperature?
- Using the output noise model and a *medium temperature* of 273 K

$$\begin{aligned} T_{\text{sky}} &= 273 (1 - G_l) \\ &= 273 (1 - 0.891) = 29.7 \text{ K } (\sim 30 \text{ K}) \end{aligned}$$

Atmospheric and rain attenuation

- Example #5
- The Ku-band satellite link has additional rain attenuation on the 11 GHz downlink of 4.0 dB
- What is the sky noise temperature?
- Adding up losses due to clear air (0.5 dB) and rain we have $G_l = 4.5$ dB and

$$\begin{aligned} T_{\text{sky}} &= 273 (1 - G_l) \\ &= 273 (1 - 0.355) = 176 \text{ K} \end{aligned}$$

Atmospheric and rain attenuation

- Example #5
- With 4.0 dB rain attenuation on downlink

$$T_{\text{sky}} = 176 \text{ K}$$

In clear air, T_{sky} was 30 K

If $T_{\text{LNA}} = 100 \text{ K}$ and we use $T_{\text{antenna}} = T_{\text{sky}}$

$$T_{\text{s rain}} = 276 \text{ K}, \quad T_{\text{s clear air}} = 130 \text{ K}$$

Increase in noise power is $10 \log (276/130)$

$$\Delta N = 3.3 \text{ dB}$$

Atmospheric and rain attenuation

- Observe that the downlink is adversely affected in two ways:
 - 1) an increase in attenuation of 4 dB ($C \downarrow 4$ dB)
 - 2) an increase in noise of 3.3 dB ($N \uparrow 3.3$ dB)
- The overall C/N in the downlink will decrease by 7.3 dB
- This may be enough to wipe out the margin and cause an outage

Atmospheric and rain attenuation

- On the downlink rain has a serious impact on the sky noise temperature at the receiver
- On the uplink with 4.0 dB rain attenuation the sky noise temperature at the satellite transponder does not change
- Why not?

Atmospheric and rain attenuation

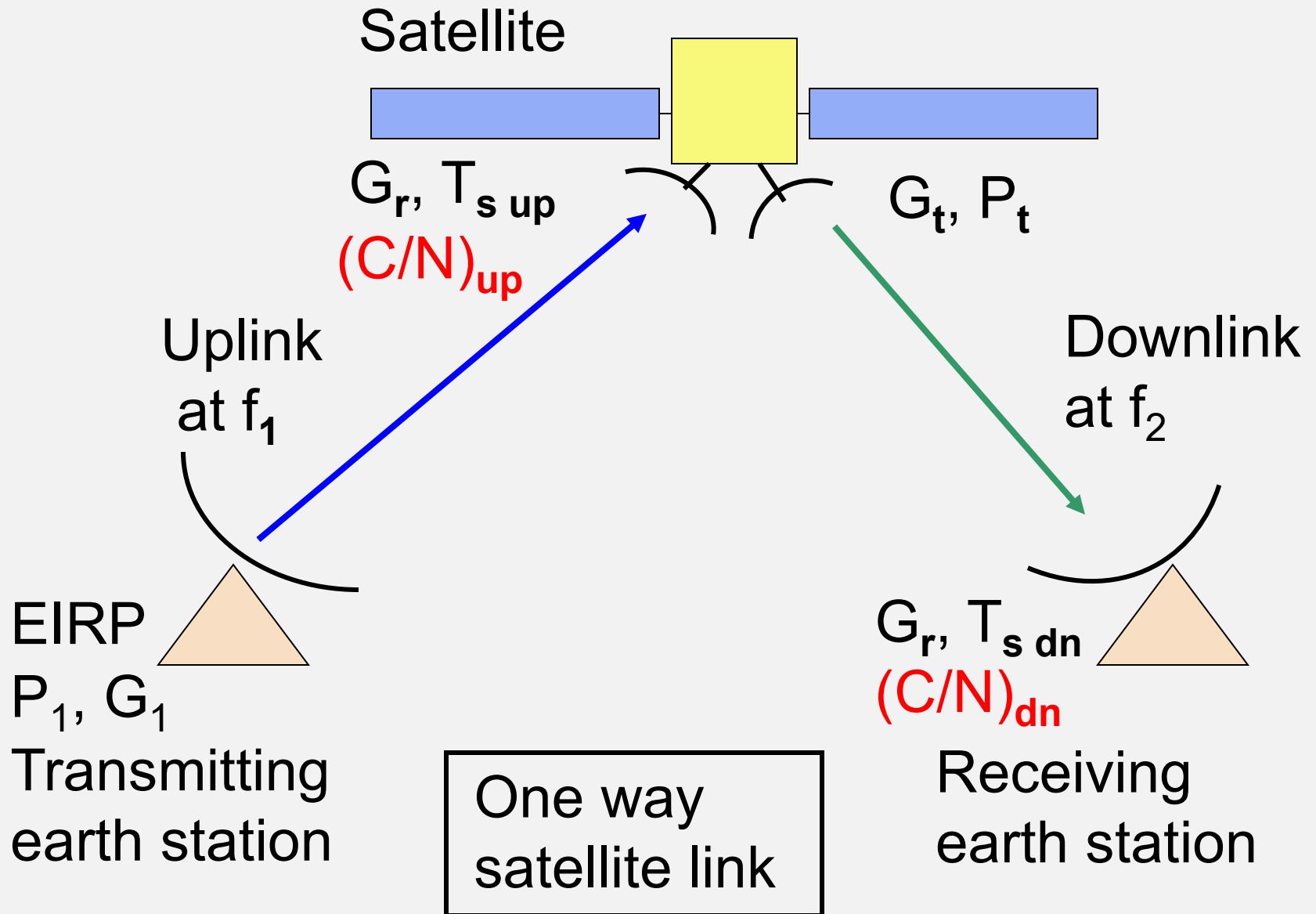
- On the downlink rain has a serious impact on the sky noise temperature at the receiver
- On the uplink with 4.0 dB rain attenuation the sky noise temperature at the satellite transponder does not change
- Since the satellite looks at “hot” earth (~ 270 K) all of the time, ‘sky’ temperature is almost constant
- Rain doesn’t change $T_{s \text{ up}}$

Atmospheric and rain attenuation

- BUT the uplink still suffers a drop in C/N of 4 dB due to the increase in direct attenuation
- Furthermore, for a linear transponder there is the same drop in the output power of the satellite transponder
- And the downlink C/N is then also affected by rain on the uplink
- How do we combine C/N on the uplink and downlink to obtain the overall (effective) C/N?

C/N on satellite links

- A satellite link consists of an *uplink* from a transmitting earth station to the satellite and a *downlink* from the satellite to a receiving earth station
- Each link has an associated (C/N) ratio, namely $(C/N)_{\text{up}}$ and $(C/N)_{\text{dn}}$



C/N on satellite links

- $(C/N)_{\text{up}}$ and $(C/N)_{\text{dn}}$ combine into an overall ratio, $(C/N)_o$, for the combined link
- $(C/N)_o$ is measured at the output of the IF amplifier or input to the demodulator at the Earth station
- $(C/N)_o$ determines the performance of the combined link
- Intuitively, it is clear that $(C/N)_o$ cannot be better, i.e., higher, than the lower of $(C/N)_{\text{up}}$ and $(C/N)_{\text{dn}}$

C/N on satellite links: Uplink

- In the transponder we have:

$$(C/N)_{\text{up}} = P_r / k T_{\text{s up}} B_n$$

- P_r is measured at the antenna terminals
- $T_{\text{s up}}$ is the system noise temperature for the transponder
- B_n is the noise bandwidth

C/N on satellite links: Uplink

- Note: B_n is the final bandwidth of the *earth station receiver* (not the transponder bandwidth)
- B_n is usually set by the bandwidth of the IF filter that precedes the demodulator
- The earth station IF filter is the ‘chokepoint’ for the link
- We can ignore noise power that falls outside the passband of this final filter before the demodulator

C/N on satellite links: Downlink

- At the Earth Station we have:

$$(C/N)_{\text{down}} = P_r / k T_{s \text{ dn}} B_n$$

- P_r is measured at the antenna terminals
- $T_{s \text{ dn}}$ is the system noise temperature for the earth station
- B_n is the noise bandwidth of the earth station receiver (IF)

C/N on satellite links

- Overall C/N ratio at the output of the earth station receiver is given by (Eqn 4.50 – text)

$$1/(C/N)_o = 1/(C/N)_{up} + 1/(C/N)_{dn}$$

- This is called the *reciprocal formula*
- Similar to the equation for the resistance of two resistors in parallel
- C/N values are linear ratios – **NOT in dB**

Adding C/N Ratios using dB

- If $(C/N)_{\text{up}} = (C/N)_{\text{dn}}$
then $(C/N)_o = (C/N) - 3 \text{ dB}$ (C/N up or dn)
- If $(C/N)_{\text{dn}} = (C/N)_{\text{up}} - 10 \text{ dB}$
then $(C/N)_o \sim (C/N)_{\text{dn}} - 0.5 \text{ dB}$
- If $(C/N)_{\text{dn}} = (C/N)_{\text{up}} - 20 \text{ dB}$
then $(C/N)_o \sim (C/N)_{\text{dn}} - 0.1 \text{ dB}$
- If $(C/N)_{\text{dn}} = (C/N)_{\text{up}} - 30 \text{ dB}$
then $(C/N)_o \sim (C/N)_{\text{dn}}$

C/N on satellite links

- Example #6

Uplink C/N = 20 dB ratio 100

Downlink C/N = 20 dB ratio 100

$$\begin{aligned} 1/(C/N)_o &= 1/(C/N)_{up} + 1/(C/N)_{dn} \\ &= 1 / 100 + 1 / 100 \\ &= 0.02 \end{aligned}$$

$$(C/N)_o = 50 \text{ or } 17.0 \text{ dB}$$

C/N on satellite links

- Example # 7

Uplink C/N = 30 dB ratio 1000

Downlink C/N = 20 dB ratio 100

$$\begin{aligned} 1/(C/N)_o &= 1/(C/N)_{up} + 1/(C/N)_{dn} \\ &= 1 / 1000 + 1 / 100 \\ &= 0.011 \end{aligned}$$

$$(C/N)_o = 90.9 \text{ or } 19.6 \text{ dB}$$

Summary

- C ($= P_r$, received signal power) is calculated using the link equation
- N ($= P_n$, noise power in receiver) is calculated using a noise model and the kTB relationship
- Rain in a satellite link path causes attenuation and, on the downlink, increases noise temperature
- $(C/N)_o$ determines the performance of the overall link via the reciprocal formula

